

Math 421

Monday, September 29

Announcements

- Homework 4 due Friday
- Grace's Drop-In Hours
 - Wednesday: 12-1 VV 311
 - Thursday: 9-11 VV B205
- MLC
 - Proof Table: M-Th 3-7:30 VV B227
 - Course Assistant: Th 4-7 Table 4

Convergence

Def: A sequence (s_n) of real numbers is said to **converge** to the real number s provided that:

For each $\epsilon > 0$ there exists a number N such that
whenever $n > N$ implies $|s_n - s| < \epsilon$.

We can think of the definition of convergence as a game:
someone gives you (a very small) $\epsilon > 0$ and you need to
find some N that makes the statement “whenever $n > N$,
it follows that $|s_n - s| < \epsilon$ ” true.

Group Activity – Divide & Conquer

Form groups of 3-6 and further split into 3 subgroups (A, B, C). Each subgroups should find the appropriate N for their specific ϵ value in each of the following examples. Then compare as a full group and answer the discussion question.

1. $s_n = \frac{1}{n}, s = 0, \epsilon = 1, \frac{1}{10}, \frac{1}{100}$

2. $s_n = \frac{1}{e^n}, s = 0, \epsilon = 1, \frac{1}{10}, \frac{1}{100}$

Q: Can you generalize what you found to obtain an expression for N given an arbitrary $\epsilon > 0$, for each example?

Recipe to show that $\lim s_n = s$

Step 0: Do scratch work to determine what N should be.

Step 1: Fix an arbitrary $\epsilon > 0$ and state the explicit expression for N , that you determined in your scratch work.

Step 2: Show that for all $n \geq N$, $|s_n - s| < \epsilon$. This often follows the algebra from your scratchwork.

Step 3: State that since $\epsilon > 0$ was arbitrary, you have proven, from the definition, that $\lim s_n = s$.

Example: Show that $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$

Discussion.

Proof.

Example: Show that $\lim_{n \rightarrow \infty} \frac{n^2+1}{n^2} = 1$

Discussion.

Activity – Write the Proof.

Activity: Show that $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} = 0$

Discussion.

Proof.

Example

Theorem. If $\lim s_n = s$, $s_n \geq 0$ for all n , and $s > 0$, then $\lim \sqrt{s_n} = \sqrt{s}$.

Discussion.

Example

Theorem. If $\lim s_n = s$, $s_n \geq 0$ for all n , and $s > 0$, then $\lim \sqrt{s_n} = \sqrt{s}$.

Proof.

Activity – Fill in the Blanks

Theorem. If $\lim s_n = s$, then $\lim 5s_n = 5s$.

Proof. Fix $\epsilon > 0$. Since $\lim s_n = s$, there exists some $N > 0$ such that: if $n > N$, then $|s_n - s| < \underline{\hspace{2cm}}$.

Assume $n > N$. Then

$$|5s_n - 5s| = \underline{\hspace{4cm}} < \epsilon.$$

Since $\epsilon > 0$ was arbitrary, $\lim 5s_n = 5s$.

Activity

Theorem. If $\lim s_n = s$ and $s_n \in [-M, M]$ for all n , then $\lim(s_n)^2 = s^2$.

Hint: $a^2 - b^2 = (a + b)(a - b)$.

Limits are Unique

Thm: If $\lim s_n = s$ and $\lim s_n = t$, then $s = t$.

Proof.

Divergence

Def: A sequence that does not converge is said to diverge.

Examples:

Activity: Write the negation of the definition of convergence.